

0.2 X 0.5
2.01 X 0.5

Note----(Specify your assumptions clearly.)

Unless given specifically in the question----

Take -- $V_{DD} = V_{CC} = 3.3V$,

For NMOS device $\mu_n C_{ox} = 140 \mu A/V^2$, $V_T = 0.7V$, $\lambda = 0.1 V^{-1}$, $V_{ov} = 0.2V$

For PMOS device $\mu_p C_{ox} = 40 \mu A/V^2$, $V_T = -0.7V$, $\lambda = 0.1 V^{-1}$, $V_{ov} = 0.2V$

For NPN/ PNP device $\beta = 100$, $V_{CE,SAT} = 0.2V$, $V_A = 100V$, $kT/q = 25mV$ (at room temp.), $I_s = 10^{-14} A$, $V_{BE,ON} = 0.6V$, $\alpha \approx 1$

NOTE:

If not specified in question -----

- Ignore γ , λ in drain current equation. Bulk of nmos connected to ground and bulk of pmos connected to V_{dd} .
- Unless specified, assume all MOSFETs are biased in the active region All symbols have the usual meaning.

Part A: CLOSED BOOK Total 2 questions in Part A.

Time: 60 minutes

Max. Marks=40

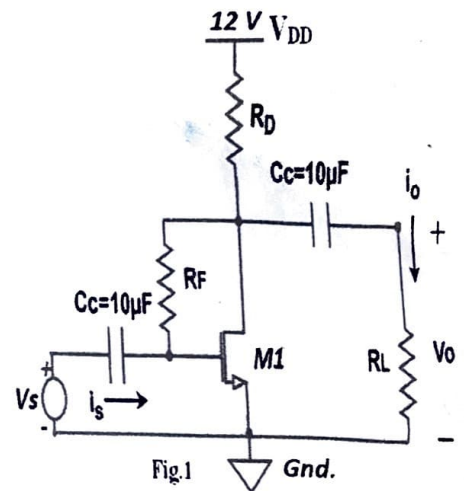
Date: 06-05-2023

Answer all the sub-parts of a question in sequence and one place only. Clearly show the procedure used to arrive at the answer for full credit. Report the answers with proper units.

- Q 1.** Consider the Amplifier circuit given in Fig.1. Following values are given for the circuit: $R_F = 100 K\Omega$, $R_D = 5 K\Omega$, $R_L = 5 K\Omega$, $K_n' = 20 \mu A/V^2$, $L = 10 \mu m$, $W = 200 \mu m$, $V_A = 100 V$, $V_T = 2 V$. Assume all the capacitors are open for DC bias and short at given signal frequency.

Determine the following:

- DC Operating point (I_{DQ} , V_{GSQ} , V_{DSQ})
- Input Impedance and Output Impedance
- Voltage gain and Current gain



[20 marks]

- Q 2. (A)** Consider the circuit of Fig. 2

Given, $|V_{tn}| = |V_{tp}| = 0.6V$, $\mu_n C_{ox} = 200 \mu A/V^2$, $\mu_p C_{ox} = 60 \mu A/V^2$.

---The channel length of all transistors = $1 \mu m$.

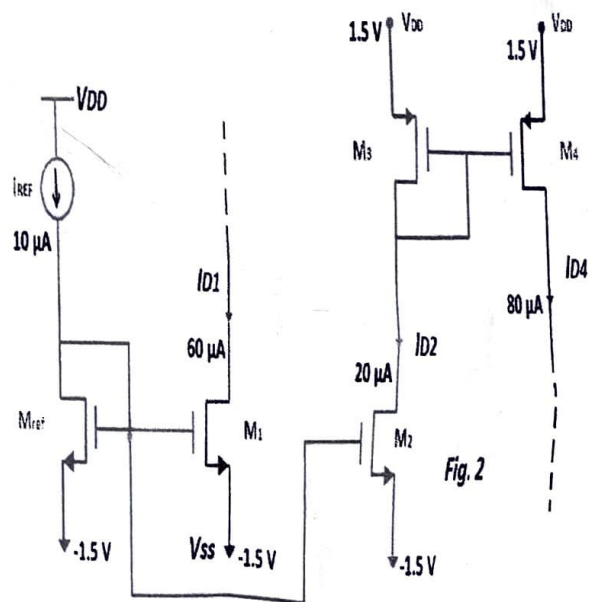
----The minimum allowed drain voltage at M_1 and M_4 are $-1.3V$ and $1.3V$, respectively.

Find the widths of M_1 , M_2 , M_4 , and M_3 .

- Q2(B)** For a p-channel long MOSFET, threshold Voltage (V_T) = $-1V$, $W = 20 \mu m$, $L = 2 \mu m$. Given, $\mu_p = 200 cm^2/V \cdot s$, $C_{ox} = 3.5 \times 10^{-7} F/cm^2$. The source and body are connected to the ground.

Calculate the drain current for following conditions----

- at $V_G = -4V$, $V_D = -1V$.
- at $V_G = -3V$, $V_D = -3V$.



[20 marks]

End of Part-A

Part-B (Open Book)

NOTE-- There are **FOUR** questions. Answer all the sub-parts of a question in sequence and one place only. Clearly show the procedure used to arrive at the answer for full credit. Report the answers with proper units.

Q1. Consider the circuit shown in Fig 1., where the applied input voltages are:

$$v_{in1} = 5\text{mV} \sin \omega t + 1.4\text{V} \text{ and } v_{in2} = 10\text{mV} \sin \omega t + 1.4\text{V}.$$

--Transistors M1, M2 has $V_{ov} = 0.2\text{V}$. The value of $I_{ss} = 90\text{uA}$

-- I_{ss} is implemented with basic current mirror with large R_{ss} .

--Assume perfect matching of transistors,

--Resistors with $R_{D1} = R_{D2} = R_D = 40\text{K}\Omega$.

a) Now, analyze the given circuit and calculate the dc voltage at nodes P and X.

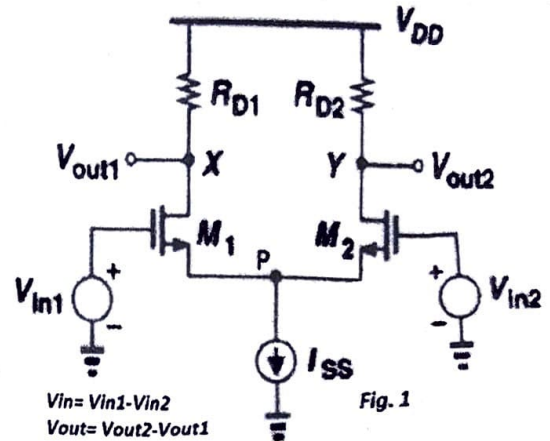
b) Calculate the differential mode gain, (A_{dm}), for the given circuit.

c) Determine common mode component of each input signal (v_{in1} and v_{in2}) and differential mode component of each input signal (v_{in1} and v_{in2})

d) Now, determine a.c value of total V_{out} tapped between v_{out1} & v_{out2} nodes. ($V_{out} = v_{out1} - v_{out2}$).

e) Finally, draw a **labelled plot** for $V_{out}(t)$ vs. time. (show both DC and AC components)

[19 marks]



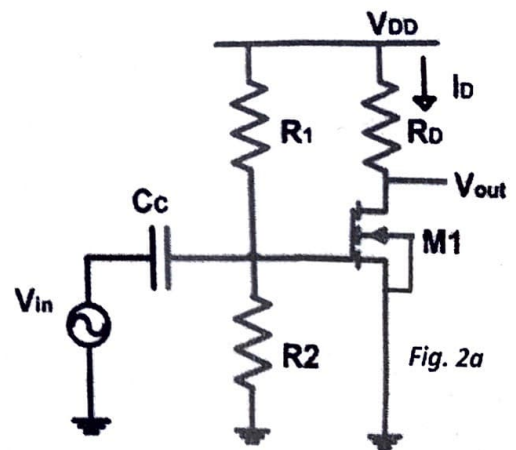
Q2. (A) Consider Fig 2a

Given: $W/L = 25$, $R_1 \parallel R_2 = 100\text{k}\Omega$, $V_{TN} = 1\text{V}$, $I_{DQ} = 2\text{mA}$, $R_D = 2.5\text{k}\Omega$

a) Draw the Load line of the given circuit and mark the maximum value of I_D and V_{DS} on the x-axis and y-axis, respectively.

b) Find the value of V_{DSQ} and V_{GSQ} in such a way that the Q point is in the **middle** of the saturation (active) region.

c) Find the value of R_1 and R_2 .



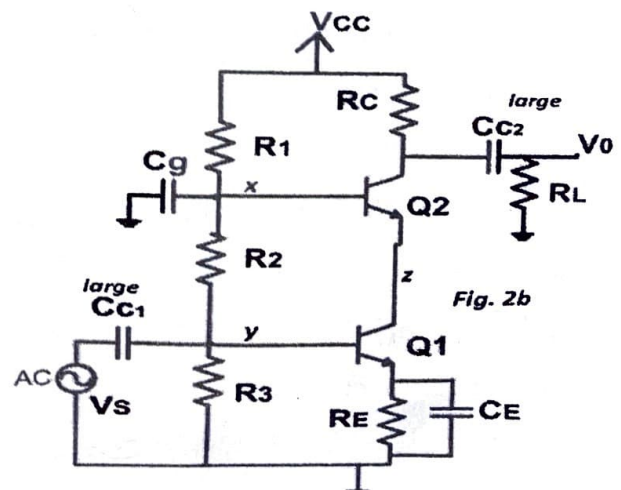
Q2 (B) Consider Fig. 2b of cascode amplifier

d) Sketch and label the small-signal (low frequency) equivalent model for the given amplifier.

(Assuming $\lambda \Rightarrow 0$)

e) Find out the gain expression of the above amplifier.

Compare the gain expression with a conventional Common emitter/base amplifier.



[Total 19 marks]

Q3. Consider the circuit given below in Fig. 3.

The Following parameters are given:

$$V_{ov} = 0.2V, \lambda = 0.01V^{-1}, C_{gs(1,2)} = 10pF,$$

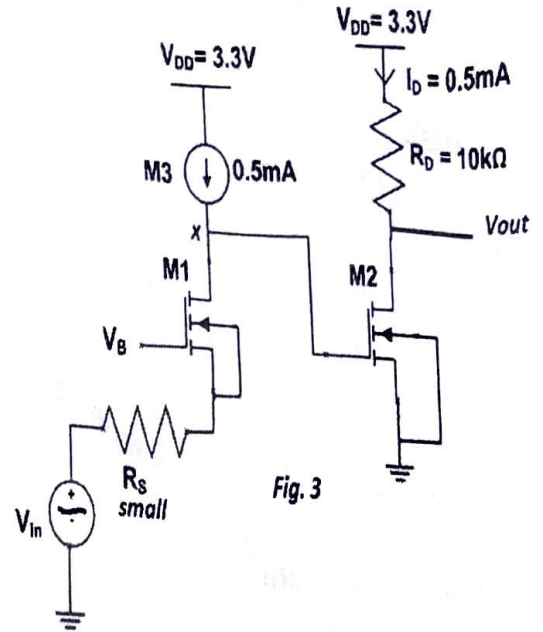
$$C_{gd(1,2)} = 1pF, C_{db(1,2)} = 2pF, C_{sb(1,2)} = 2pF \text{ and}$$

$$R_s \ll 1/g_m$$

Ignore the capacitances of the basic current mirror current source M3

- Sketch and label the high frequency model of the amplifier.
- Identify the **number** of dominant poles and zeros of the amplifier.
- Calculate the frequency (in rad./s) of the **dominant poles and zeros** of the amplifier.
- Calculate the **phase** of the amplifier at $\omega = 10K \text{ rad/s}$.
- Plot and label the **Bode magnitude response** of the amplifier (Use corner plot) qualitatively. Mark the values of low frequency voltage gain (v_{out}/v_{in}) and the dominant poles and zeros along with UGB on it.
- Calculate the unity gain frequency (**UGB in rad./s**)

[17 marks]



Q4. Consider the circuit given in Fig. 4.

The bias current $I_5 = 25\mu A$, $I_{SS} = 200\mu A$.

For all devices take, $V(\text{overdrive}) = 0.2V$,

$\lambda = 0.01V^{-1}$ Assume load capacitance at output is

$C_L = 1nF$, R_{sig} is small

$R_p = 1K\Omega$, $R_F = 1M\Omega$, $R_S = 1K\Omega$

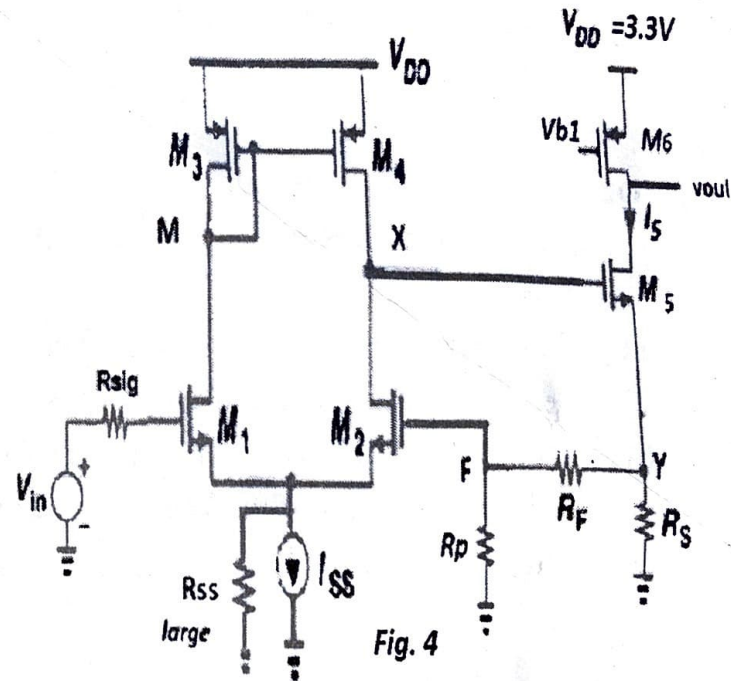
Neglect body effect in calculations.

- Identify the **type of feedback** in this amplifier.

Hence determine **type** of input and output signal of feedback amplifier

- Determine the **open loop** parameters A_o , R_{out} at low frequency
- Determine **closed loop** parameters: A_{of} , R_{inf} , R_{outf} , feedback factor (β) at low frequency
- Assuming that **output node pole** is dominant pole, determine the **-3dB frequency**, and **UGB** (unity gain bandwidth) in both **open loop** and **closed loop** condition.

Also determine **gain-crossover frequency** (G_x) in closed loop mode.



*** End of Part-B ***

(15 marks)